

Fourier and Wavelet-Based Extraction of Wave Parameters from Tanager Hyperspectral Imagery

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1. Project Overview

This project investigates the potential of hyperspectral satellite imagery for extracting wave information required for wave-based bathymetric inversion in coastal environments. The study area is Santa Cruz, Galápagos, Ecuador, where obtaining detailed bathymetric observations through conventional field surveys is logistically challenging and costly.

The central hypothesis is that spatial wave patterns visible in satellite imagery contain information related to the underlying bathymetry. If wave characteristics such as wavelength, propagation direction, and energy can be reliably extracted from imagery, these observations can subsequently be incorporated into wave-dispersion relationships to estimate water depth.

The project therefore focuses on developing a complete image-processing pipeline for wave-parameter extraction, including spectral classification, detection of wave patterns, and characterisation of wave phenomena such as refraction, diffraction, and interference.

2. Data and Processing

The experiment uses Tanager hyperspectral imagery, including visual imagery (ortho_visual.tif), surface-reflectance hyperspectral data (ortho_sr_hdf5.h5), and cloud/quality masks (ortho_beta_udm.tif). Tanager provides more than 400 spectral bands covering approximately 400–2,500 nm, with approximately 5 nm spectral spacing and around 30 m spatial resolution.

The Python workflow was developed using open-source scientific libraries including NumPy, Rasterio, SciPy, scikit-image, h5py, scikit-learn, Matplotlib, PyWavelets, and scikit-image.

The processing sequence consists of seven main stages:

a) Data Loading and Pre-processing

- Loading the Tanager orthorectified visual product and hyperspectral surface-reflectance data (HDF5 format).
- Extracting spectral bands in the visible and near-infrared regions (blue, green, red, NIR, SWIR1) using wavelength matching.
- Loading the beta cloud mask for quality filtering.

b) Supervised Classification (Random Forest)

- Interactive ROI selection using a custom graphical interface with keyboard controls (W, L, C, N, F, S, V) for seven land-cover classes: (WATER, LAND, CLOUD, NO_DATA, FRESH_WATER, SHADOW, VEGETATION)
- Training a Random Forest classifier (300 trees, Gini impurity, balanced subsampling) using five spectral bands.

- Generating a 7-class classification map and individual binary masks (GeoTIFF).
- Validation using a 20% hold-out set, yielding overall accuracy and class-wise metrics.

c) Water Mask Extraction

- Isolating the WATER class (class 0) from the classification map to restrict wave analysis to the water domain.

d) Wave Detection (Wavelet + FFT)

- Extracting 64×64 windows centred on sampled water pixels (up to 3,000 samples).
- Computing 2D Fast Fourier Transform (FFT) for each window and calculating band-limited energy ratios in three frequency bands (0.05–0.15, 0.15–0.30, 0.30–0.50 cycles/pixel).
- Computing wavelet features using Daubechies-4 decomposition (approximation and detail coefficients).
- Training a Random Forest classifier on these features to generate a probabilistic wave-presence map.
- Producing a spatially smoothed wave-probability map (GeoTIFF).

e) Wave Parameter Extraction

- For pixels with high wave probability (top 90th percentile), extracting a 64×64 window and computing the 2D FFT.
- Locating the dominant spectral peak (excluding the DC component) in the frequency domain.
- Deriving physical parameters: (Spatial frequency (cycles/pixel); Wavelength (m) using pixel size (30 m); Propagation direction (degrees, 0° = North); Energy ratio (relative energy of the dominant peak))
- Saving results as a CSV file (wave_parameters.csv) with coordinates and probability.

f) Wave Phenomenon Analysis (Refraction, Diffraction, Interference)

- Computing local metrics for each detected wave point:
 - Orientation gradient: measures spatial change of wave direction (indicates refraction due to depth or current variations).
 - Curvature: measures bending of wave fronts along propagation (indicates diffraction around obstacles).
 - Circular variance of orientation: measures directional dispersion in a neighbourhood (indicates interference between crossing wave trains).
- Generating maps of these metrics as GeoTIFFs (refraction_gradient.tif, diffraction_curvature.tif, interference_variance.tif).

- Visualising zones of refraction (colour-coded arrows), diffraction (yellow points), and interference (red crosses) on the original RGB image.

g) Visualisation and Reporting

- Generating a rose diagram (directional histogram) of wave orientations with wavelength and energy distributions.
- Creating a comprehensive figure with original RGB, enhanced wave image, probability map, orientation map, and wavelength map.
- Overlaying wave-direction vectors on the original image, with linear wave-theory estimates (period calculation from wavelength: $T = \sqrt{2\pi L/g}$).
- Saving all outputs in a dedicated output directory (wave_detection_output/).

3. Technical Innovation

The methodological contribution is the integration of spectral classification, spatial-frequency analysis, and wave-phenomenon characterisation into a unified workflow for extracting quantitative wave descriptors from hyperspectral imagery.

In the spatial domain, wave patterns can be difficult to characterise objectively because their appearance is affected by illumination, image contrast, background variability, and other coastal features. In the frequency domain, however, approximately periodic wave structures produce concentrated spectral energy at characteristic spatial frequencies.

The FFT provides a mechanism for separating dominant wave structures from broader spatial variability and extracting their orientation and frequency characteristics. The wavelet transform complements this by capturing localised textural features that may not be fully represented by global Fourier analysis.

The additional analysis of refraction, diffraction, and interference distinguishes this workflow from simpler wave-detection approaches. By quantifying spatial gradients of direction, curvature of wave fronts, and directional variance, the method provides insight into the physical processes affecting wave propagation in the coastal zone—information that is valuable for understanding depth-induced transformations.

The Random Forest classification step is also innovative because it enables the isolation of the water domain using supervised learning, which is more robust than threshold-based indices (e.g., NDWI) in complex coastal environments with mixed pixels, shadows, and varying water types.

4. Relevance to Bathymetric Inversion

The ultimate objective is to use remotely observed wave characteristics as inputs to a wave-dispersion-based bathymetric inversion.

The workflow follows this conceptual chain:

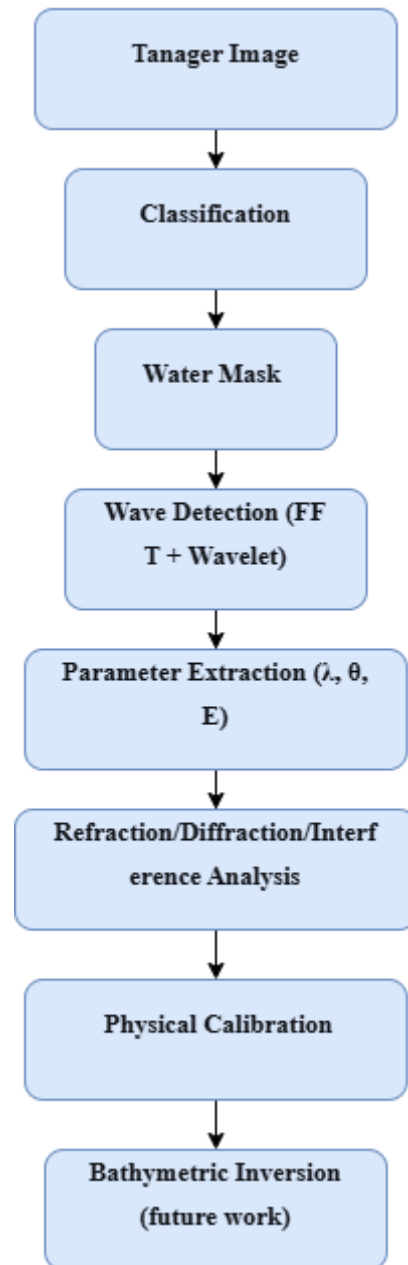


Figure 1 Tanager Hyperspectral Wave Parameter Extraction Workflow. Source: Author, 2026

This is particularly relevant for shallow coastal environments where wave transformation is influenced by depth. Instead of requiring direct measurements of depth at every location, the method seeks to exploit the physical relationship between wave propagation and bathymetry.

At the current stage, the project establishes the image-processing and wave-parameter extraction component. The extracted wavelength values are expressed in metres using the known pixel size (30 m) and can therefore be directly introduced into a physical wave model. The wave period is estimated from the linear dispersion relation in deep water ($T = \sqrt{2\pi L/g}$).

The next methodological step is to incorporate wave-period and environmental information, apply the full dispersion relationship in intermediate or shallow water, and evaluate the resulting bathymetric estimates against independent observations.

5. Role of Tanager

Tanager was fundamental to this proof of concept because it provides hyperspectral observations with substantially finer spectral sampling than conventional multispectral satellite imagery. Its coverage of the visible through shortwave-infrared spectrum and approximately 5 nm spectral spacing provide an opportunity to investigate coastal-water signatures at a much more detailed spectral scale.

For this project, the most valuable features were:

Hyperspectral data cube: providing both spectral and spatial information for classification and wave detection.

Surface-reflectance product: atmospheric correction already incorporated, delivering analysis-ready data.

Orthorectified visual product: enabling georeferenced visualisation and direct overlay of wave vectors.

Cloud and quality masks: facilitating data quality filtering.

The combination of spectral water identification with spatial-frequency analysis was enabled by having access to both the visual RGB and the full hyperspectral cube.

6. Expected Impact

A successful wave-based bathymetric inversion framework could provide a complementary source of bathymetric information for coastal areas where conventional surveys are expensive, spatially sparse, or difficult to conduct.

Potential users include:

- Coastal and oceanographic researchers
- Hydrographic and mapping organisations
- Environmental monitoring agencies
- Coastal-zone managers
- Numerical-modelling groups
- Researchers developing coastal digital twins
- Organisations requiring spatially continuous information on shallow-water morphology

The potential benefit is not intended to replace high-accuracy hydrographic surveying. Instead, satellite-derived information could provide a scalable complementary dataset for reconnaissance, model initialisation, temporal monitoring, and identification of areas requiring targeted field surveys.

7. Current Status and Future Development

The current implementation demonstrates a complete prototype from Tanager image acquisition to wave-pattern detection, parameter extraction, and wave-phenomenon analysis. The workflow successfully integrates:

- Supervised classification for water-domain isolation (7 classes).
- FFT and wavelet-based wave detection with probabilistic mapping.

- Physical parameter extraction (wavelength, orientation, energy).
- Analysis of refraction, diffraction, and interference.
- Comprehensive visualisation and reporting.

All processing is implemented in Python and is fully reproducible. The output products include GeoTIFF maps, CSV tables with quantitative parameters, and high-resolution figures for interpretation.

Future development will focus on four main components:

- **Physical calibration:** refine wavelength estimation using precise pixel-size calibration and image-geometry corrections.
- **Wave physics:** integrate wave period, propagation direction, and full dispersion relationship (including intermediate/shallow-water effects).
- **Bathymetric inversion:** estimate depth from remotely observed wave characteristics and validate against independent data.
- **Validation and scalability:** test the method across multiple coastal environments, acquisition dates, and wave conditions.

Additional Tanager observations would make it possible to test the method under different wave conditions, water optical properties, coastal morphologies, and seasonal states. Expanding the study beyond Santa Cruz would also provide an important test of transferability.

Conclusion

This project demonstrates a complete proof-of-concept workflow for converting hyperspectral satellite observations into quantitative wave information relevant to coastal bathymetry. By combining spectral classification, Fourier and wavelet analysis, and wave-phenomenon characterisation, the workflow establishes a pathway from satellite imagery to the wave parameters required for physical bathymetric inversion.

The broader objective is to investigate whether hyperspectral Earth observation can provide a scalable and cost-effective complementary source of bathymetric information for remote and data-sparse coastal environments.

References

- Almar, R., & others. (2019).** Water depth inversion from satellite dataset based on wave celerity and wavelength. *HAL BRGM*.
- Bergsma, E. W. J., & Almar, R. (2018).** The influence of optical satellite resolutions on the ability to resolve waves. *Remote Sensing of Environment*, 210, 140–151.
- Bergsma, E. W. J., & Almar, R. (2020).** Coastal coverage of ESA + Sentinel missions for the retrieval of bathymetry and waves. *Remote Sensing*, 12(7), 1156.
- Breiman, L. (2001).** Random forests. *Machine Learning*, 45(1), 5–32.
- Carbon Mapper / Planet Labs. (2024).** *Advanced Hyperspectral Imaging from Orbit: Achievements and Challenges from the First Year of Tanager-1*. Digital Commons USU.

- Casal, G., Harris, P., Monteys, X., Hedley, J., & Cahalane, C. (2021).** Pan-European satellite-derived coastal bathymetry—Review, user needs and future services. *Frontiers in Marine Science*, 8, 740505.
- Chen, K. S., & others. (2008).** Internal wave detection and parameter estimation from SAR images based on a novel Radon transform method. *IEEE Xplore*.
- Danilo, C., & Binet, R. (2018).** Estimation of shallow water bathymetry using linear wave dispersion theory. *IEEE Xplore*.
- Danilo, C., & Binet, R. (2019).** Monitoring beach topography and nearshore bathymetry using spaceborne remote sensing: A review. *Remote Sensing*, 11(19), 2283.
- ESA. (2025).** *Tanager Constellation Updates*. LPS25.
- Gao, J. (2009).** Bathymetric mapping by means of remote sensing: methods, accuracy and limitations. *Progress in Physical Geography*, 33(1), 103–116.
- Garello, R., & others. (2002).** Wavelet transform of SAR images for internal wave detection and orientation. *IEEE Xplore*.
- Gupta, R., & others. (2025).** Improving land cover mapping in riverine environments with machine learning and spectral indices. *ISPRS Annals*, X-4/W2-2025, 45–52.
- Holman, R., & Stanley, J. (2007).** The history and technical capabilities of Argus. *Coastal Engineering*, 54(6-7), 477–491.
- Jawak, S. D., & Luis, A. J. (2015).** A comprehensive review on satellite-derived bathymetry. *Journal of the Indian Society of Remote Sensing*, 43(3), 421–435.
- Kumar, A., & others. (2025).** Spatio-temporal flood inundation dynamics using Random Forest classification. *Science of the Total Environment*, 950, 175221.
- Laigle, R., Almar, R., Bergsma, E. W. J., & others. (2021).** Outlier correction in remote sensing retrieval of ocean wave wavelength and application to bathymetry. *Remote Sensing*, 13(19), 3901.
- Li, X., & others. (2020).** Assessment of wind direction estimation methods from SAR images with Radon transform enhancement. *Remote Sensing*, 12(22), 3785.
- Liu, Y., & others. (2022).** Evaluation of feature extraction algorithms for oceanic internal waves based on nighttime detection data. *IEEE Xplore*.
- Lyons, A. P., & others. (1997).** Wavelet analysis in SAR ocean image profiles for internal wave detection and wavelength estimation. *IEEE Xplore*.
- Misra, A., & Ramakrishnan, B. (2015).** *Extraction of coastal ocean wave characteristics using remote sensing and computer vision technologies*. CSIR-NIO Technical Report.
- Poupardin, A., & others. (2017).** Estimation of shallow water bathymetry using linear wave dispersion theory on a single Resourcesat-2 LISS-IV image. *IEEE Xplore*.
- Poupardin, A., Idier, D., de Michele, M., & Raucoules, D. (2016).** Water depth inversion from a single SPOT-5 dataset. *IEEE Transactions on Geoscience and Remote Sensing*, 54(4), 2329–2342.
- Rodriguez, G., & Garello, R. (1998).** Internal wave detection and location in SAR images using wavelet transform. *IEEE Xplore*.

Singh, P., & others. (2025). Land use land cover mapping using Random Forest and multispectral satellite imagery in Google Earth Engine. *IEEE Xplore*.

USGS. (2026). *System Characterization Report on Tanager*. USGS Publications Warehouse.

Yang, C., & others. (2015). Retrieval of ocean wavelength and wave direction from SAR image based on Radon transform. *IEEE Xplore*.